

Progress Since Customer Day 5

Since Customer Day 5, the team focused on consolidating the project into a deployment-ready and maintainable solution. While the previous phase concentrated mainly on feature completion and user experience improvements, Customer Day 6 centred on preparing the platform for final delivery and handover. The key additions completed were conversion to a cloud native architecture, integration of background music to enhance the interactive experience, and preparation of the documentation package for handover. Together, these changes strengthen both the technical robustness of the system and the completeness of the final product.

Customer Day Timeline

Customer Day 2: The Pivot & Foundation

- The Shift: Originally, the team proposed a "Virtual Pet" system. Based on client feedback (SCS), you pivoted to a conversational, RPG-style interface inspired by *Pokémon*.
- Planning: Established the core technical stack: Next.js, Tailwind CSS, and Supabase.
- User Stories: Defined the first high-priority stories: Gamified Survey (frontend) and Insight Dashboard (backend/admin).
- Risk Assessment: Identified "Gamification Fatigue" and accessibility for elderly users as primary concerns.

Customer Day 3: Concept to Prototype

- Infrastructure: Set up the GitHub repository and deployed the first live web application on Render.
- The Storyline: Introduced the "Guardian Cell" narrative users help defend the body from cancer cells by providing feedback.
- Design Evolution: * Replaced the QR scanner with a PIN-entry alternative for better accessibility.
 - Transitioned the "Events Page" into an "About Us" section with direct links to SCS social media.
- Security: Implemented Anonymous UUIDs via browser cookies to prevent duplicate submissions while maintaining privacy.

Customer Day 4: Real-Time Integration

- Backend Connectivity: Successfully linked the dashboard to the live Supabase database, enabling real-time analytics for the client.
- Separation of Concerns: Implemented Multi-Event Support, allowing the client to manage different feedback forms (e.g., different booths or dates) independently.

- Feature Expansion: Added management controls for questions and event settings directly within the admin interface, reducing the need for developer intervention.

Customer Day 5: Professionalization & Logic

- Advanced Logic: Finalized the "Cell A vs. Cell B" branching narrative. The ending (who is captured) now depends on the user's specific responses, making the feedback feel impactful.
- Role-Based Access (RBAC): Built a secure login for Superadmins (full access) and Admins (assigned events only).
- Admin Console: Added a dedicated assignment console for Superadmins to manage their team's access levels.
- Localization: Added full support for Singapore's four national languages (English, Chinese, Malay, Tamil).

Customer Day 6: The Polished Finale

- UI/UX Refinement: Replaced all placeholder assets with official SCS branding and logos.
- Advanced Analytics: Upgraded the dashboard with Pie Charts for visual sentiment analysis and a Booth Completion Rate tracker.
- Direct QR Generation: Integrated a tool inside the dashboard to generate event QR codes on the fly.
- Code Quality: Completed a full refactoring pass to ensure the project is maintainable and adheres to the "Professional" standard of the module.

Implemented Enhancements

Containerised Application

The application has now been fully containerised to ensure that the full system can run in a consistent and reproducible environment. Separate containers are defined for the dashboard, feedback form, and backend services, with Docker Compose coordinating the complete stack.

The backend is split into three micro-services, each responsible for one area of the system.

Events Service

This service manages all event-related functionalities, including the creation and modification of events, quest management, and the assignment of administrative users. Access control is enforced by validating users' authentication tokens and roles (e.g., admin or superadmin). In addition, the database implements Row-Level Security (RLS) as a secondary safeguard, ensuring that users can only access or modify data for which they have explicit permissions.

Feedback Service

This service is responsible for managing feedback questions and collecting participant responses. Upon submission, the service records the feedback response along with all associated answers through a single database operation using a remote procedure call (RPC). This approach ensures transactional integrity, meaning that either all data is successfully stored or none is saved, thereby eliminating the risk of incomplete or inconsistent records.

Analytics Service

The system computes engagement metrics, such as survey completion rates, the frequency of skipped questions, and trends in responses over the past 11 days. These computations are performed on the server to offload processing from the client device and to ensure that analytical workloads do not impact the performance of the feedback submission process.

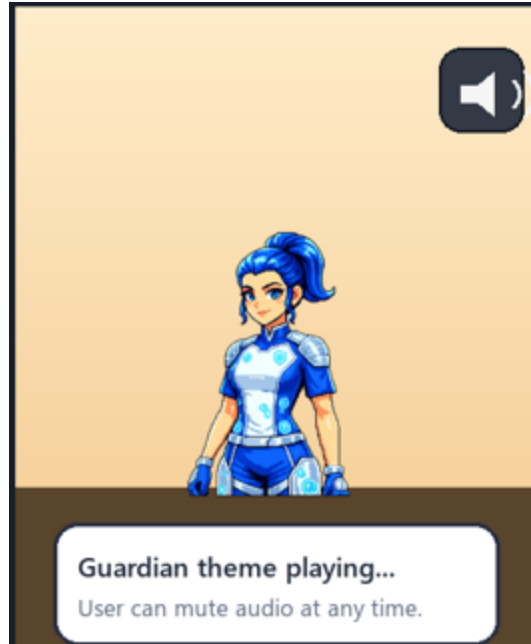
Frontend Applications

The system is divided into two distinct frontend applications. The Dashboard (port 3000) is designed for organisers and provides administrative functionalities, while the Feedback Form (port 3001) is dedicated to participants and focuses on response submission. This separation ensures a clear distinction between administrative and user-facing features, improving maintainability, security, and user experience.

Background Music Integration

To further enhance the RPG-style feedback experience, background music was added across key parts of the application. Different scenes now include audio elements that better support the tone of the storyline and make the user journey feel more immersive. This helps transform the application from a purely functional feedback system into a more engaging and polished interactive experience.

The addition of audio was implemented with usability in mind. Users are still given control over the experience through mute and unmute functionality, ensuring that the feature remains optional and does not disrupt accessibility or comfort. As a result, the background music contributes to atmosphere and presentation while still respecting user preference.



Documentation Handover

A major focus of Customer Day 6 was the preparation of the documentation handover package. The team consolidated the technical and operational information required for future developers, maintainers, or client-side stakeholders to understand and continue working with the platform after project completion. This includes documentation for system overview, architecture, local setup, environment configuration, service responsibilities, deployment flow, testing, observability, and troubleshooting.

The handover package also includes operational runbooks for common failure scenarios, such as service downtime and elevated error rates. These documents provide step-by-step guidance for identifying problems, checking service health, reviewing logs, and applying recovery measures. This ensures that the project is handed over not only as a working codebase, but as a maintainable system with sufficient supporting knowledge for ongoing use and support.

Handover Items Checklist

Item	Descriptions
GitHub Repository	Current codebase

Documentations	System architecture, setup, and usage documentation
Hosting & Domain Ownership	Ownership of hosting platforms and domain services
Database	Core database including schema

Handover Documentation Overview

The handover documentation is intended to give the recipient both technical understanding and practical ownership of the project after delivery.

System-Level Understanding

At the system level, it explains the purpose of the platform and how the major parts of the solution work together. This includes the public-facing user experience, the administrative interface, the supporting backend services, and the database layer. The purpose of this documentation is to help the customer understand the structure of the system rather than receiving only a set of project files without context.

Operational-Level Guidance

At the operational level, the documentation covers setup requirements, environment configuration, local execution, deployment flow, infrastructure arrangement, validation steps, rollback approach, testing coverage, observability, and troubleshooting. This part of the handover is important because it enables the customer to run, deploy, and maintain the system without dependence on the development team.

Database Ownership and Structure

The handover also includes database-related documentation and exported database structure information. Its purpose is to ensure that the client receives control not only over the running system, but also over the underlying data model. This supports future maintenance, migration, and recovery.

Runbooks and Operational Support

In addition, we included dedicated runbook material for operations support. These runbooks are more specific than the general technical documentation. Their purpose is to guide the customer through concrete actions during incidents, such as when a service becomes unavailable or when error rates increase unexpectedly. This is valuable because it improves readiness for real operational issues after deployment.

Summary of Handover Documentations

Overall, the documentation being handed over explains the system, supports setup and deployment, preserves database ownership, and provides practical incident-response guidance through runbooks. Together, these materials make the handover more complete, usable, and aligned with customer expectations.

Conclusion

Customer Day 6 focused on finalising the platform for delivery. By containerising the application, enhancing the user experience through background music, and preparing a structured documentation handover, the team improved not only the polish of the product but also its deployment readiness and long-term maintainability. These final additions complete the project in a way that supports both demonstration and future continuation.